

Variation of Parameters

ANSWER KEY



The Wronskian

- 1 For $y_1 = e^x$, $y_2 = e^{2x}$ (solutions of $y'' - 3y' + 2y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
 $y_1' = e^x$, $y_2' = 2e^{2x}$. $W = e^x(2e^{2x}) - e^x(e^{2x}) = 2e^{3x} - e^{3x} = e^{3x}$.
- 2 For $y_1 = \cos x$, $y_2 = \sin x$, compute W .
 $y_1' = -\sin x$, $y_2' = \cos x$. $W = \cos x \cos x - (-\sin x) \sin x = \cos^2 x + \sin^2 x = 1$.

The variation of parameters formula

- 3 State the variation of parameters formula for a particular solution of $y'' + p(x)y' + q(x)y = g(x)$, given independent solutions y_1, y_2 of the homogeneous equation with Wronskian W .

$$y_p = -y_1 \int \frac{y_2 g}{W} dx + y_2 \int \frac{y_1 g}{W} dx.$$

A forcing term with no undetermined-coefficients trial form

- 4 Solve $y'' + y = \tan x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters.
Here $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$, $g = \tan x$. Since $\int \cos x \tan x dx = \int \sin x dx = -\cos x$ and $\int \sin x \tan x dx = \int \sec x dx - \int \cos x dx = \ln |\sec x + \tan x| - \sin x$, we get
 $y_p = -\cos x [\ln |\sec x + \tan x| - \sin x] + \sin x (-\cos x) = -\cos x \ln |\sec x + \tan x|$. General solution:
 $y = C_1 \cos x + C_2 \sin x - \cos x \ln |\sec x + \tan x|$.

A repeated root, with forcing dividing by x

- 5 Solve $y'' - 2y' + y = \frac{e^x}{x}$ ($x > 0$) using variation of parameters.
 $r^2 - 2r + 1 = (r - 1)^2 = 0$, so $y_1 = e^x$, $y_2 = xe^x$, and $W = e^{2x}$. With $g = e^x/x$: $\frac{y_2 g}{W} = 1$ so $\int = x$; $\frac{y_1 g}{W} = \frac{1}{x}$ so $\int = \ln x$. Then $y_p = -e^x(x) + xe^x(\ln x) = xe^x(\ln x - 1)$. Absorbing the $-xe^x$ term into the arbitrary constant on y_2 , the general solution is $y = C_1 e^x + C_2 x e^x + x e^x \ln x$.

Comparing with undetermined coefficients

- 6 Solve $y'' - y = e^{2x}$ by variation of parameters, then check the particular solution using undetermined coefficients.

$y_1 = e^x$, $y_2 = e^{-x}$, $W = -2$. With $g = e^{2x}$: $\frac{y_2 g}{W} = -\frac{1}{2}e^x$, integral $-\frac{1}{2}e^x$, $\frac{y_1 g}{W} = -\frac{1}{2}e^{3x}$, integral $-\frac{1}{6}e^{3x}$. So $y_p = -e^x(-\frac{1}{2}e^x) + e^{-x}(-\frac{1}{6}e^{3x}) = \frac{1}{2}e^{2x} - \frac{1}{6}e^{2x} = \frac{1}{3}e^{2x}$. Check by undetermined coefficients: try Ae^{2x} (no resonance, since 2 is not a root of $r^2 - 1 = 0$): $4A - A = 1 \Rightarrow A = \frac{1}{3}$. Matches. General solution:
 $y = C_1 e^x + C_2 e^{-x} + \frac{1}{3}e^{2x}$.

The Wronskian for complex-root solutions

- 7 For $y_1 = e^{-x} \cos 2x$, $y_2 = e^{-x} \sin 2x$, compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
 $y_1' = -e^{-x}(\cos 2x + 2 \sin 2x)$, $y_2' = e^{-x}(2 \cos 2x - \sin 2x)$. Expanding $W = y_1 y_2' - y_1' y_2$, the cross terms cancel and $W = e^{-2x}(2 \cos^2 2x + 2 \sin^2 2x) = 2e^{-2x}$.

Setting up variation of parameters (without full integration)

- 8 For $y'' + 4y = \sec 2x \tan 2x$, with $y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$, set up the two integrals in the variation-of-parameters formula (don't evaluate).

$$y_p = -\cos 2x \int \frac{\sin 2x \sec 2x \tan 2x}{2} dx + \sin 2x \int \frac{\cos 2x \sec 2x \tan 2x}{2} dx.$$

A polynomial forcing term, cross-checked against undetermined coefficients

- 9 Solve $y'' + y = x$ using variation of parameters ($y_1 = \cos x$, $y_2 = \sin x$, $W = 1$), and verify it matches the undetermined-coefficients answer $y_p = x$.

$\int x \sin x dx = -x \cos x + \sin x$; $\int x \cos x dx = x \sin x + \cos x$. So
 $y_p = -\cos x(-x \cos x + \sin x) + \sin x(x \sin x + \cos x) = x \cos^2 x + x \sin^2 x = x$ — matching undetermined coefficients exactly (direct substitution: $0 + x = x$ confirms $y_p = x$ is correct).

A repeated root with a forcing term dividing by x-squared

- 10 Solve $y'' + 2y' + y = \frac{e^{-x}}{x^2}$ ($x > 0$) using variation of parameters.

$r^2 + 2r + 1 = (r + 1)^2 = 0 \Rightarrow y_1 = e^{-x}$, $y_2 = x e^{-x}$, $W = e^{-2x}$. With $g = e^{-x}/x^2$: $\frac{y_2 g}{W} = \frac{1}{x}$, integral $\ln x$;

$\frac{y_1 g}{W} = \frac{1}{x^2}$, integral $-\frac{1}{x}$. So $y_p = -e^{-x} \ln x + x e^{-x} \left(-\frac{1}{x}\right) = -e^{-x}(\ln x + 1)$. Absorbing the constant term,

the general solution is $y = C_1 e^{-x} + C_2 x e^{-x} - e^{-x} \ln x$.

Summary: choosing undetermined coefficients vs. variation of parameters

- 11** For $y'' - y = e^{2x} \tan x$, explain why variation of parameters is required rather than undetermined coefficients.

$\tan x$ is not a polynomial, exponential, sine/cosine, or a product of these — the finite list of forcing types undetermined coefficients can guess a trial form for. Variation of parameters works for any continuous forcing $g(x)$, so it's required here.

One more Wronskian computation

- 12** For $y_1 = e^{3x}$, $y_2 = e^{-x}$ (solutions of $y'' - 2y' - 3y = 0$), compute the Wronskian $W = y_1 y_2' - y_1' y_2$.
 $y_1' = 3e^{3x}$, $y_2' = -e^{-x}$. $W = e^{3x}(-e^{-x}) - 3e^{3x}(e^{-x}) = -e^{2x} - 3e^{2x} = -4e^{2x}$.
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A trigonometric forcing term via variation of parameters

- 13** Solve $y'' + y = \sec x$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$ using variation of parameters ($y_1 = \cos x$, $y_2 = \sin x$, $W = 1$).
 $\int \sin x \sec x dx = \int \tan x dx = -\ln |\cos x|$; $\int \cos x \sec x dx = \int 1 dx = x$. So
 $y_p = -\cos x(-\ln |\cos x|) + \sin x(x) = \cos x \ln |\cos x| + x \sin x$. General solution:
 $y = C_1 \cos x + C_2 \sin x + \cos x \ln |\cos x| + x \sin x$.
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One final setup exercise

- 14** For $y'' - 4y' + 4y = \frac{e^{2x}}{x^3}$ ($x > 0$), with $y_1 = e^{2x}$, $y_2 = xe^{2x}$, $W = e^{4x}$, set up the two integrals in the variation-of-parameters formula (don't evaluate).

$$y_p = -e^{2x} \int \frac{xe^{2x} \cdot e^{2x}/x^3}{e^{4x}} dx + xe^{2x} \int \frac{e^{2x} \cdot e^{2x}/x^3}{e^{4x}} dx = -e^{2x} \int \frac{dx}{x^2} + xe^{2x} \int \frac{dx}{x^3}.$$

A second forcing term with no elementary trial form

- 15** Solve $y'' - 2y' + y = \frac{e^x}{1+x^2}$ using variation of parameters ($y_1 = e^x$, $y_2 = xe^x$, $W = e^{2x}$).

$$\frac{y_2 g}{W} = \frac{x}{1+x^2}, \text{ integral } \frac{1}{2} \ln(1+x^2); \frac{y_1 g}{W} = \frac{1}{1+x^2}, \text{ integral } \arctan x. \text{ So}$$
$$y_p = -e^x \cdot \frac{1}{2} \ln(1+x^2) + xe^x \arctan x. \text{ General solution: } y = C_1 e^x + C_2 x e^x - \frac{1}{2} e^x \ln(1+x^2) + x e^x \arctan x.$$

A closing Wronskian and setup pair

- 16** For $y_1 = e^{-2x}$, $y_2 = e^{5x}$ (solutions of $y'' - 3y' - 10y = 0$), compute $W = y_1 y_2' - y_1' y_2$.
 $y_1' = -2e^{-2x}$, $y_2' = 5e^{5x}$. $W = e^{-2x}(5e^{5x}) - (-2e^{-2x})(e^{5x}) = 5e^{3x} + 2e^{3x} = 7e^{3x}$.

- 17** For $y'' + 9y = \csc 3x$, with $y_1 = \cos 3x$, $y_2 = \sin 3x$, $W = 3$, set up the two integrals in the variation-of-parameters formula (don't evaluate).

$$y_p = -\cos 3x \int \frac{\sin 3x \csc 3x}{3} dx + \sin 3x \int \frac{\cos 3x \csc 3x}{3} dx.$$

A final full worked example

- 18** Solve $y'' + 4y = \sec 2x \tan 2x$ using variation of parameters (continuing the setup from an earlier question, with $y_1 = \cos 2x$, $y_2 = \sin 2x$, $W = 2$).

$$\int \frac{\sin 2x \sec 2x \tan 2x}{2} dx = \int \frac{\tan^2 2x}{2} dx = \frac{1}{2} \left(\frac{\tan 2x}{2} - x \right) \text{ (using } \tan^2 \theta = \sec^2 \theta - 1 \text{)}.$$

$$\int \frac{\cos 2x \sec 2x \tan 2x}{2} dx = \int \frac{\tan 2x}{2} dx = -\frac{1}{4} \ln |\cos 2x|. \text{ So}$$

$$y_p = -\cos 2x \cdot \frac{1}{2} \left(\frac{\tan 2x}{2} - x \right) + \sin 2x \left(-\frac{1}{4} \ln |\cos 2x| \right), \text{ which simplifies (using } \cos 2x \tan 2x = \sin 2x \text{) to}$$

$$y_p = -\frac{1}{4} \sin 2x + \frac{1}{2} x \cos 2x - \frac{1}{4} \sin 2x \ln |\cos 2x|.$$

A last Wronskian check

- 19** For $y_1 = x$, $y_2 = x^2$ (solutions of the equation $x^2 y'' - 2xy' + 2y = 0$ on $x > 0$), compute $W = y_1 y_2' - y_1' y_2$.
 $y_1' = 1$, $y_2' = 2x$. $W = x(2x) - 1(x^2) = 2x^2 - x^2 = x^2$.